

Analytic Continuation, Zeta Regularization, and Ontic Structural Realism: Stabilizing Singularities through Mathematical Structure

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Abstract

Physical singularities—such as those in general relativity and quantum field theory have long been taken as indicators of theoretical breakdowns. This paper examines analytic continuation and zeta regularization as mathematical techniques that stabilize such divergences by extending physical quantities into domains of well-defined meaning. It argues that this process exemplifies a deeper ontological stabilization of physical theory: where structure, rather than substance, carries the explanatory weight. Drawing on the framework of Ontic Structural Realism (OSR), the paper contends that zeta-regularized quantities, including the analytically continued spectral zeta function $\zeta_A(s)$ of the Laplace-Beltrami operator, model an ontology in which mathematical continuation restores coherence to otherwise ill-defined regions of theory. Thus, mathematical regularization is not merely a calculational device but a structural act revealing how the formal architecture of mathematics stabilizes physical ontology at the limits of knowledge.

Keywords: analytic continuation, zeta regularization, singularities, Ontic Structural Realism, spectral geometry, mathematical ontology

1. Introduction

The use of analytic continuation and zeta regularization in modern physics extends far beyond technical renormalization. At its core lies a metaphysical insight: singularities may mark the limits of description, not the limits of reality. When a divergent quantity is continued analytically, the mathematics does not erase the singularity but re-expresses it within a broader structural framework. This paper argues that such stabilization through structure embodies the central tenet of Ontic Structural Realism (OSR): that the world's fundamental ontology consists in relational structure rather than in individual entities.

Section 2 develops the mathematical foundations of zeta regularization as a rigorous analytic continuation technique. Section 3 translates this formalism into the philosophical vocabulary of OSR, identifying structural continuity across divergent regimes. Section 4 then synthesizes these perspectives in cosmology, interpreting the Big Bang and inflationary

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dynamics as analytic rather than ontological boundaries. The concluding section articulates the broader significance of this framework for the philosophy of physics.

2. Mathematical Structure: Zeta Regularization and Analytic Continuation

Zeta regularization provides a consistent means of assigning finite values to divergent series and integrals. For a positive self-adjoint operator A with spectrum $\{\lambda_i\}$, the spectral zeta function

$$\zeta_A(s) = \sum_i \lambda_i^{-s}$$

converges for $\text{Re}(s)$ sufficiently large and admits analytic continuation to a meromorphic function on $\mathbb{C}$. Physical quantities—Casimir energies, partition functions, and effective actions—are often defined through the derivative $\zeta'_A(0)$ or the analytically continued value $\zeta_A(0)$.

Hawking (1977) first applied this formalism to quantum fields in curved spacetime, showing that regularization preserves diffeomorphism invariance. Elizalde (1995) later developed a comprehensive account connecting spectral geometry to renormalization in curved space. Mathematically, the continuation replaces the divergence at a pole with a finite structural invariant determined by the analytic residue, thus stabilizing the otherwise undefined expression.

3. Ontic Structural Realism and Analytic Ontology

Ontic Structural Realism (OSR) asserts that structure is ontologically primary. Ladyman (1998) and Rickles (2008) advanced this position within the philosophy of physics, arguing that entities are merely nodes in a web of relations described by mathematical form. Within this view, analytic continuation exemplifies how mathematical relations, not intrinsic objects, determine the persistence of ontology through theoretical transformation.

Regularization thus becomes a metaphysical act of structural preservation: it ensures that a theory remains coherent even where its naive formulation fails. The continuation of $\zeta_A(s)$ across a pole does not eliminate the singularity but reinterprets it as an analytic boundary—an expression of relational persistence across formal discontinuities. In this sense, analytic continuation embodies the OSR principle that reality is defined by invariant structure rather than by the specific numerical values of divergent quantities.

4. Zeta-Structural Cosmology

4.1. The Cosmological Singularity as Analytic Boundary

The classical singularity at $t = 0$ in Friedmann–Lemaître–Robertson–Walker (FLRW) cosmology represents, mathematically, a pole in the spacetime curvature scalar. Within the zeta-regularized framework, this pole can be treated analogously to a divergent spectral sum: rather than a breakdown of ontology, it marks an analytic boundary beyond which continuation defines a regularized domain. The Big Bang is therefore not an ontological rupture but an analytic boundary in the cosmic function $t \mapsto g_{\mu\nu}(t)$.

4.2. Heat-Kernel Cosmology

The heat-kernel expansion,

$$K(t) = \text{Tr } e^{-tA} \sim \sum_{n=0}^{\infty} a_n t^{(n-d)/2},$$

provides a bridge between spectral geometry and cosmology. As $t \rightarrow 0$, divergent terms correspond to ultraviolet behavior near the Planck epoch. Zeta regularization defines $\zeta_A(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} K(t) dt$ and extends it analytically in s , effectively continuing the physical description through the singular limit $t \rightarrow 0$. Thus, the analytic continuation in s mirrors the physical extrapolation toward the origin of cosmic time.

4.3. Inflation as Analytic Expansion

Inflation can be interpreted as the physical realization of analytic coefficients within this continuation. Each coefficient a_n in the heat-kernel series corresponds to a geometric invariant contributing to spacetime dynamics. The rapid exponential expansion of the early universe then appears as the unfolding of these analytic structures into the metric fabric of spacetime. Inflation becomes the manifestation of analytic structure in cosmological evolution.

4.4. Finite Vacuum Energy and the Cosmological Constant

In zeta regularization, vacuum energy is rendered finite through analytic continuation, yielding

$$E_{\text{vac}} = \frac{1}{2} \hbar \zeta_A(-\frac{1}{2}).$$

This finite structural quantity replaces the divergent sum over zero-point modes. Interpreted philosophically, the cosmological constant problem is reframed: fine-tuning is not a physical coincidence but an analytic necessity. The regularized universe possesses a structurally determined vacuum energy fixed by the analytic properties of $\zeta_A(s)$, aligning physical stability with mathematical consistency.

4.5. Ontological Continuity Beyond the Singularity

By extending the analytic domain of $\zeta_A(s)$, the continuation defines an ontological continuity beyond singular boundaries. The singularity is absorbed into the structure of the function itself—analogous to how complex analysis unifies apparent discontinuities within a single analytic surface. Reality, under this interpretation, is analytically connected even where classical physics predicts rupture.

4.6. The Zeta-Regularized Cosmos

The zeta-regularized cosmos is one whose existence is guaranteed by analytic structure. Each physical singularity corresponds to a boundary of definition rather than a boundary of being. Ontology and analysis coincide: the universe persists not through substance but through the continuity of mathematical form.

5. Conclusion and Philosophical Synthesis

This study has argued that analytic continuation and zeta regularization embody a metaphysical principle of ontological stabilization. The mathematical act of continuing a divergent function mirrors the philosophical act of extending ontology beyond apparent limits. Within the OSR framework, zeta regularization becomes a paradigm of structural realism: mathematics reveals the continuity of reality through its own analytic coherence.

Three contributions follow. First, the framework provides a structural resolution of singularities, treating them as analytic boundaries. Second, it reinterprets vacuum energy and the cosmological constant as finite structural invariants. Third, it unites the metaphysical stance of OSR with the formal machinery of regularization, showing that structural relations constitute the enduring core of physical ontology.

Future work may extend this analytic-ontological framework to black-hole singularities, quantum-gravity regularization, or other dualities where analytic structure mediates between divergent regimes. The broader implication is that the mathematics of continuation reveals a deep principle of reality's resilience: existence is secured not by substance, but by the analytic unity of structure.

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Declarations

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